

5.9.11

EE25BTECH11057 - Rushil Shanmukha Srinivas

Question: Sumit is 3 times as old as his son. Five years later, he shall be two and a half times as old as his son. How old is Sumit at present ?

Solution: Let Age of son = x

Age of Father = y = 3x

$$3x - y = 0 \quad (0.1)$$

Five years later, father shall be two and a half times as old as his son, So

$$y + 5 = \frac{5}{2}(x + 5) \quad (0.2)$$

$$5x - 2y = -15 \quad (0.3)$$

The system of equations formed is :

$$3x - y = 0 \quad (0.4)$$

$$5x - 2y = -15 \quad (0.5)$$

Writing it in the matrix form,

$$\begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -15 \end{pmatrix} \quad (0.6)$$

Forming the augmented matrix to solve the system of equations,

$$\left(\begin{array}{cc|c} 3 & -1 & 0 \\ 5 & -2 & -15 \end{array} \right) \quad (0.7)$$

Using Gaussian Elimination,

$$\left(\begin{array}{cc|c} 3 & -1 & 0 \\ 5 & -2 & -15 \end{array} \right) \xleftrightarrow{R_2 \rightarrow 3R_2 - 5R_1} \left(\begin{array}{cc|c} 3 & -1 & 0 \\ 0 & -1 & -45 \end{array} \right) \quad (0.8)$$

Using back substitution we get,

$$-y = -45 \quad (0.9)$$

$$y = 45 \quad (0.10)$$

$$3x - y = 0 \quad (0.11)$$

$$3x = 45 \quad (0.12)$$

$$x = 15 \quad (0.13)$$

The solution for the system of equations is :

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15 \\ 45 \end{pmatrix} \quad (0.14)$$

Therefore the present age of father = $y = 45$

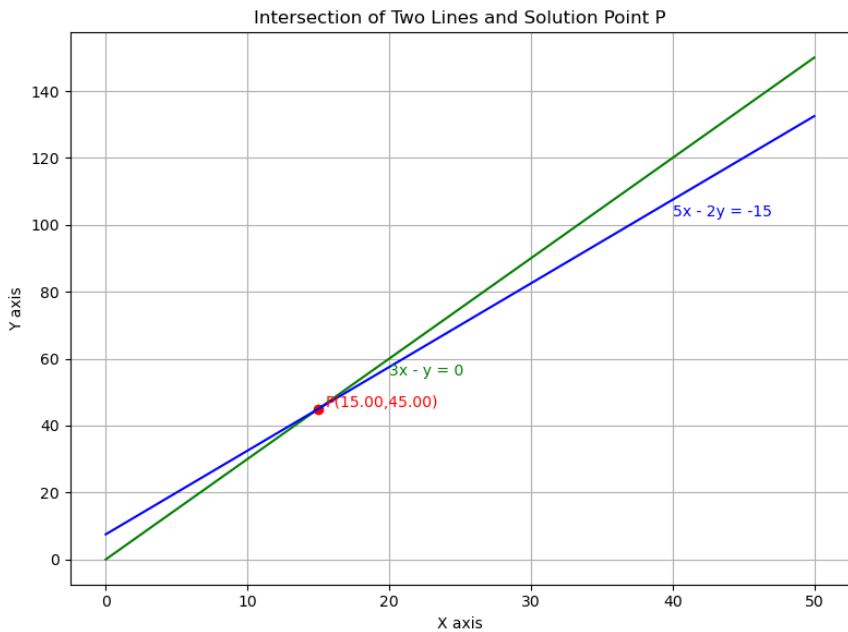


Fig: Representation of Lines and Point of Intersection